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TP Modul 7

main.cpp

```cpp
#include "stack.h"

int main() {
    int input, totalJumStack;
    stack S;

    createStack_103032400084(S);

    for (int i = 0; i < MAXSTACK; i++) {
        cout << "input angka ke-" << i + 1 << ": ";
        cin >> input;

        push_103032400084(S, input);

        if (i == MAXSTACK-1) {
            cout << "stack sudah penuh!" << endl;
        }
    }

    cout << endl;
    totalJumStack = sumStack_103032400084(S);
    cout << "hasil penjumlahan elemen dalam stack: " << totalJumStack << endl;

    return 0;
}
```

stack.cpp

```cpp
#include "stack.h"

void createStack_103032400084(stack &S) {
    S.top = 0;
}

bool isEmpty_103032400084(stack S) {
    return (S.top == 0);
}

bool isFull_103032400084(stack S) {
    return (S.top == MAXSTACK);
}

void push_103032400084(stack &S, infotype x) {
    if (isFull_103032400084(S) == false) {
        S.info[S.top] = x;
        S.top++;
    }
}

infotype pop_103032400084(stack &S) {
    infotype x;

    S.top--;
    x = S.info[S.top];

    return x;
}

int sumStack_103032400084(stack S) {
    int total = 0;

    while (!isEmpty_103032400084(S)) {
        total = total + pop_103032400084(S);
    }

    return total;
}

```

stack.h

```cpp
#define STACK_H_INCLUDED
#include <iostream>

using namespace std;

const int MAXSTACK = 10;

typedef int infotype;

struct stack {
    infotype info[MAXSTACK];
    int top;
};

void createStack_103032400084(stack &S);
bool isEmpty_103032400084(stack S);
bool isFull_103032400084(stack S);
void push_103032400084(stack &S, infotype x);
infotype pop_103032400084(stack &S);
int sumStack_103032400084(stack S);

#endif // STACK_H_INCLUDED
```

output

```
input angka ke-1: 1
input angka ke-2: 2
input angka ke-3: 3
input angka ke-4: 4
input angka ke-5: 5
input angka ke-6: 6
input angka ke-7: 7
input angka ke-8: 8
input angka ke-9: 9
input angka ke-10: 10
stack sudah penuh!

hasil penjumlahan elemen dalam stack: 55
```